iPerturb: A Biologically Informed and Explainable Framework for Context-Specific Single-Cell Perturbation Response Modelling

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Abstract. Building an interpretable model that predicts genome-wide transcriptional responses to genetic perturbations remains a grand challenge in biology. Mechanistic kinetic models are difficult to scale to the genome, whereas deep neural network–based models often lack interpretability. To bridge this gap, we propose iPerturb, a biologically informed and explainable framework for context-specific single-cell perturbation response modelling. iPerturb first integrates five tiers of regulatory and association evidence into a cell-line-specific gene regulatory network (GRN); then propagates a perturbation through it with a learnable transfer function that follows the Hill function of gene regulation, whose Chebyshev-basis form turns propagation into matrix multiplication; and finally explains each prediction through the named neighbours that support it. Evaluated on held-out CRISPRi targets in two human cell lines (699 in K562 and 743 in RPE1), the context-specific GRN raises the centred correlation of the predicted response over a linear baseline by +0.093 [+0.077, +0.108] and +0.082 [+0.059, +0.106] (95% intervals), whereas degree-preserving rewired and data-derived graphs give none. The Hill-form operator reproduces exact Hill propagation to 3.5×10^{-5} and, on a dense graph, lowers peak memory six- to eight-fold while making edge-specific learnable shapes feasible. On the curated signed core, which reaches 5.8–7.1% of the perturbed genes, Hill-form propagation matches but does not exceed linear propagation.

Keywords: gene expression prediction · genetic perturbation · gene regulatory network · biologically informed neural network · Hill function · interpretable AI

1 Introduction

Deciphering how cells transcriptionally respond to perturbations remains a central challenge in biology. Addressing it requires models that are predictive, scalable to the genome, and mechanistically interpretable [3]. Mechanistic kinetic models do not scale, as measuring kinetic parameters experimentally is expensive, limiting them to simple systems [10,9]. Deep neural network–based methods such as GEARS, scGPT, and STATE [15,19,4,11,1] scale to genome-wide

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prediction but require large training sets and offer little interpretability, and recent benchmarks show that they do not consistently surpass simple linear baselines [2]. Biologically informed neural networks (biNNs) aim to combine both via architectural priors, but existing biNNs mainly predict higher-level phenotypes [7,14] rather than genome-wide expression responses to genetic perturbations.

In this paper, we introduce iPerturb, a biNN-based framework for predicting genome-wide expression changes induced by genetic perturbations, built on two ideas. As illustrated in Figure 1, iPerturb takes a cell line and a genetic perturbation as input and works in three steps. First, cell-line context is used to query public regulatory resources and construct a *context-specific gene regulatory network* (GRN) from five evidence tiers, keeping signed regulation and unsigned association evidence strictly apart. Second, the perturbation is propagated through the GRN by a *learnable transfer function that follows the Hill function of gene regulation* [23,5], with the perturbed gene clamped at every update; because the Hill curve of every regulator is expressed in a Chebyshev basis, propagation becomes a matrix multiplication that runs efficiently on accelerators, a form related to basis-function expansions in learnable-activation networks [13,12]. Third, every prediction is explained by the named neighbours and edges that support it, and the explanation is tested by ablation.

We applied iPerturb to CRISPRi perturbations in two human cell lines, K562 and RPE1, predicting the mean response of held-out target genes. Our results show that (i) the context-specific GRN carries predictive signal for unseen perturbations that degree-preserving rewired and purely data-derived graphs do not, and that this signal survives the removal of response magnitude; (ii) the Hill-form operator reproduces exact Hill propagation to within 10^{-5} while the matrix-multiplication form lowers memory six- to eight-fold on dense graphs and makes edge-specific learnable shapes feasible; and (iii) on the curated signed core, which reaches only a small fraction of targets, Hill-form propagation matches but does not exceed linear propagation. Here, *interpretable* is meant mechanistically: each predicted change decomposes into contributions from named neighbours and edges of the curated network, rather than from opaque latent factors; we do not claim that the explanation paths are validated molecular mechanisms.

2 Data and Methods

2.1 Datasets

Public gene-regulation resources were curated from four complementary source types: (i) transcription factor (TF)–target interactions from TRRUST [8], CollecTRI [16] and OmniPath [21] (120,312 unique directed pairs, 61,215 of them with a defensible sign); (ii) enhancer–gene and TF-binding annotations from GeneHancer v5.26 [6]; (iii) protein–protein association pairs from STRING [20] filtered at combined score ≥ 700; and (iv) co-expression pairs from COXPRESdb [17]. Perturbation data are the genome-scale CRISPRi Perturb-seq screens of Replogle et al. [18] in K562 (chronic myelogenous leukaemia) and RPE1 (retinal pigment

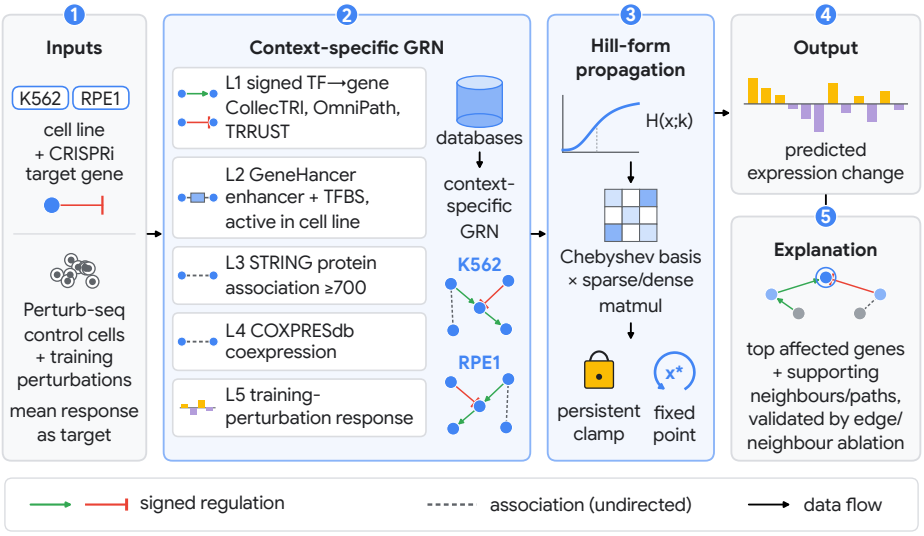


Fig. 1. Overview of the iPerturb framework. (1) A cell line and a CRISPRi target are the inputs; the response is the perturbation-conditioned mean expression change. (2) A context-specific GRN is assembled from five evidence tiers (L1 curated signed regulation; L2 enhancers and binding sites active in the cell line; L3 protein association; L4 co-expression; L5 training-perturbation response), with association evidence kept undirected and unsigned. (3) The perturbation is propagated by a learnable transfer function of Hill form, evaluated in a Chebyshev basis as a matrix product, with the perturbed gene clamped. (4) The predicted expression change is (5) explained by supporting neighbours and edges, tested by ablation. Graphs and bars are illustrative.

epithelium) cells. Starting from the raw UMI counts, each cell was normalised to 10^4 counts and log1p-transformed; the response of a perturbation p is the *mean-level* difference $\Delta_p = \overline{x}_p - \overline{x}_0$ between the mean log-normalised expression of its cells and of the control cells of the same gem groups (batch-matched), computed on the stored gene axis (8,563 genes in K562, 8,749 in RPE1). Perturbations with fewer than 20 cells and targets absent from the gene axis were removed, leaving 1,835 (K562) and 1,928 (RPE1) targets. The data are single-cell, but the predicted endpoint is the perturbation-conditioned mean response; cell-resolved generation is outside the scope of this work.

2.2 Constructing the context-specific GRN

iPerturb integrates evidence into a five-tier hierarchy ordered by strength (Level 1: curated, signed TF–target interactions; Level 2: GeneHancer enhancer–gene links whose enhancers are active in the target cell line and whose TF-binding sites match the TF; Level 3: STRING associations, added as undirected, unsigned pairs; Level 4: COXPRESdb co-expression, top partners per gene; Level 5: signed

edges read from training perturbations only). Association evidence (Levels 3–4) is *never* represented as a signed or directed regulatory arrow: it is stored as an undirected edge and used only by the association channel of Section 2.3. Cross-tier duplicates were resolved in favour of the lowest-numbered tier, and within-tier sign conflicts defaulted to unsigned, so an unknown sign is never promoted to activation. The graph is built separately for every cell line: Level 2 depends on the enhancers and TF-binding sites active in that line, and Level 5, when used, is fitted on the training perturbations of that line alone.

2.3 Learnable propagation that follows the Hill function

Hill-form regulatory propagation. Let $x_i \in (\varepsilon, 1-\varepsilon)$ be the regulatory activity of gene i and $x_i^0 = \sigma(z_i)$ its baseline, where z_i is the standardised control expression (a fixed monotone map, so the graph cannot be bypassed by a learned encoder). A regulator j passes its activity through a modified Hill transfer function

$$H(x; k, n) = \frac{k^n \left[1 - (1-x)^n\right]}{(1-x)^n + k^n}, \qquad H(0) = 0,\ H(1) = 1, \tag{1}$$

(the TF4 form of the logic-ODE literature [23,5]) with $n = 2$ and one learnable $k_j \in [0.05, 1]$ *per regulator*, shared by all of its outgoing edges. Signed messages are baseline-centred, so an unperturbed cell produces exactly zero response:

$$x_i^{t+1} = (1-\eta)\, x_i^t + \eta\, \sigma\Big(\operatorname{logit} x_i^0 + \sum_{j \to i} w_{ji} \big[H(x_j^t; k_j) - H(x_j^0; k_j)\big]\Big), \tag{2}$$

with $w_{ji} = \rho\, s_{ji}\, c_{ji}\, \mathrm{softplus}(a) / \max\{1, \sum_{j'} |\cdot|\}$, where $s_{ji} \in \{-1, +1\}$ is the curated sign, c_{ji} the evidence confidence, and ρ, a are learnable scalars. A CRISPRi knockdown of gene p is a *persistent clamp*, $x_p^t = (1-\zeta) x_p^0$ at every iteration, where ζ is the median knockdown of the training perturbations. The fixed point x^* is found by damped iteration ($\eta = 0.5$, 24 steps) and the predicted response is $\hat{\Delta}_g = d_g\,(x_g^* - x_g^0)$ with a non-negative diagonal decoder d_g; no dense $G \times G$ decoder is used, so every prediction decomposes over named edges.

Matrix-multiplication form. Per-edge Hill evaluation requires gathering x_j for every edge and scattering messages, which maps poorly to accelerators. We therefore approximate each regulator's Hill curve by a Chebyshev expansion of degree r,

$$H(x; k_j) \approx \sum_{b=0}^{r} a_b(k_j)\, T_b(2x - 1), \tag{3}$$

whose coefficients $a_b(k)$ are least-squares fitted to Eq. (1) on a logarithmic grid of k and linearly interpolated in $\log k$, so that k_j stays learnable and differentiable. With $\mathbf{h} = \hat{H}(x) - \hat{H}(x^0) \in \mathbb{R}^{B \times G}$ for a batch of B perturbations, the aggregation in Eq. (2) is one sparse (or dense) matrix product $\mathbf{h}\,\mathbf{W}$, with $\mathbf{W}_{ji} = w_{ji}$, instead of an edge-wise gather and index-add. The same fixed-point equations are solved; only the arithmetic changes (equivalence is verified in Section 3.3).

Association channel. Most perturbed genes are not curated TFs. Let $\mathbf{G} \in \mathbb{R}^{G \times k}$ be the top-k gene loadings of the centred training responses, and $\hat{A}$ the row-normalised adjacency of the (undirected) context-specific GRN. The embedding of a perturbed gene t is smoothed over its neighbours, $e_t = (1-\alpha)\mathbf{G}_t + \alpha\,(\hat{A}\mathbf{G})_t$, and the response is $\hat{\Delta}_t = \bar{\Delta} + e_t\,\mathbf{W}_e\,\mathbf{G}^\top$ with a ridge-fitted $\mathbf{W}_e$; k and α are chosen on the validation perturbations from fixed grids. For $\alpha = 0$ this is the standard linear low-rank baseline, so the channel nests it.

SCC-aware execution. Feedback makes Eq. (2) cyclic. For a frozen graph we compute strongly connected components once, order their condensation topologically, restrict the computation to the nodes reachable from the perturbed gene, settle acyclic singletons with a single forward update and iterate feedback components locally with their upstream inputs fixed. The predictions are those of the global solver (Section 3.3); this is an execution strategy, not a different model.

2.4 Evaluating iPerturb

For each cell line, targets were split into training, validation and test perturbations (70/15/15%) in three folds stratified only by properties known without response data (TF status, control-expression tertile, curated out-degree bin); no target is shared across the splits of a fold. All graph-derived quantities, loadings, knockdown efficacy and hyper-parameters use training (and validation) perturbations only. Because the mean response of a perturbation is dominated by a response shared across perturbations [22], the primary metric is the Pearson correlation of Δ *centred* on the training-mean response (ρ_c), computed per perturbation without the perturbed gene and averaged over perturbations; we also report the same correlation after projecting the training-mean direction out of both profiles (ρ_r, which removes response magnitude), the top-20 centred correlation (ρ_{20}), the mean normalised rank of the correct perturbation among all test perturbations (Cent.; 0.5 is chance) and the area under the precision–recall curve for differentially expressed genes ($|\text{Welch } z| > 4$, cell level, no biological replicates, prevalence-dominated by the shared response). Intervals are 95% bootstrap intervals over unique held-out targets (each target averaged over the folds in which it is tested); paired differences use the same bootstrap. We compare against no change, the training-mean response, a ridge model on a low-rank gene embedding [2], GEARS [19] trained on the same folds (Section 3.2), a two-layer graph convolutional network on the same GRN, and control GRNs with degree-preserving rewired edges or a graph learned from the training data alone. The full protocol was fixed in a hashed pre-registration file before fitting, and later analyses are labelled as such in the repository.

3 Results

3.1 iPerturb builds cell-line-specific GRNs with a small signed core

The GeneHancer tier is where context enters the network. Restricted to enhancers active in each line and to TFs whose binding sites match, the tier contains 645,570 TF–target pairs for K562, 335,734 for HepG2 and 80,414 for Jurkat, yet the edge sets overlap little (Jaccard index 0.08 K562–HepG2, 0.07 K562–Jurkat, 0.05 HepG2–Jurkat), and TF out-degrees agree only moderately across lines (Spearman $\rho = 0.31$ K562–HepG2 to 0.71 K562–Jurkat). The RPE1 tier is empty in this release because no RPE1-labelled TF-binding annotation is available to match, so RPE1 relies on the other tiers. The curated signed core is small relative to the screen. Only 5.8% (K562) and 7.1% (RPE1) of the perturbed genes have a signed out-edge, and 4.4% and 4.7% reach at least 100 genes. Where signed edges exist they are informative: on training perturbations, the sign of a curated edge predicts the direction of the knockdown response of its target in 80% (K562; 109 responsive edges) and 68% (RPE1; 222) of cases ($p < 10^{-7}$, binomial). The core is therefore reliable but narrow, which motivates the second, association channel.

3.2 The context-specific GRN improves prediction of held-out perturbations

Because the mean response is dominated by a component shared across perturbations, raw correlation is saturated by the training mean (0.26 in K562 and 0.49 in RPE1), and we therefore report centred correlations (Figure 2, Table 1). The smoothing head of iPerturb, in which a perturbed gene's embedding is averaged over its neighbours in the context-specific GRN, improves on the linear low-rank baseline by +0.093 [+0.077, +0.108] (K562) and +0.082 [+0.059, +0.106] (RPE1) in ρ_c, and by +0.083 [+0.069, +0.097] and +0.071 [+0.059, +0.084] after removing the shared response direction (ρ_r), so the gain is not a change in response magnitude. The same model on a degree-preserving rewired GRN gains nothing (the real GRN is better by +0.097 [+0.081, +0.113] and +0.089 [+0.066, +0.113]), and neither does a graph learned from the training data alone with the same degree sequence (the GRN is better by +0.091 [+0.076, +0.107] and +0.080 [+0.057, +0.104]). Zeroing each training row's own-target entry before the decomposition leaves the gain unchanged (+0.098 [+0.084, +0.113], +0.089 [+0.066, +0.113]). The curated regulatory tier adds nothing beyond STRING in this channel (+0.003 [-0.001, +0.007] and +0.002 [-0.006, +0.011] for L1+L3 relative to STRING alone); the signal comes from association evidence, i.e. from the response of functionally related genes; the case with the largest gain (Section 3.4) is supported by ribosomal-protein neighbours. Two simple heads on the same GRN do at least as well: a graph convolutional network (B5) and the mean training response of a gene's neighbours that are themselves training targets, which is better than the smoothing head by +0.029 [+0.019, +0.038] and +0.047 [+0.032, +0.063] (GCN) and by +0.047 [+0.035, +0.060] and +0.050

Table 1. Test results on held-out CRISPRi targets. Unique-target means over three folds (bold: best per column). ρ_c: centred correlation; ρ_r: after removing the training-mean direction; ρ_{20}: top-20 centred correlation; Cent.: mean normalised centroid rank; AUPRC: differentially expressed genes (|Welch z| > 4). Intervals for all differences are given in Table 2.

K562 (699 held-out targets) Model	ρ_c	ρ_r	ρ_{20}	Cent.	AUPRC
No change (B0)	0.096	0.000	0.116	0.503	0.099
Train mean (B1)	0.000	0.000	0.000	0.000	0.031
Linear low-rank (B2)	0.145	0.139	0.275	0.641	0.110
Smoothing, rewired GRN	0.141	0.134	0.268	0.633	0.110
Smoothing, data-kNN graph	0.147	0.141	0.277	0.640	0.112
Smoothing, STRING only	0.235	0.223	0.430	0.735	0.132
iPerturb: smoothing head	0.238	0.221	0.434	0.737	0.132
iPerturb: GCN head (B5)	0.267	0.240	0.454	0.755	0.140
iPerturb: neighbour-mean head	**0.285**	**0.276**	**0.511**	**0.791**	**0.162**
RPE1 (743 held-out targets) Model	ρ_c	ρ_r	ρ_{20}	Cent.	AUPRC
No change (B0)	0.115	0.000	0.050	0.498	**0.245**
Train mean (B1)	0.000	0.000	0.000	0.000	0.054
Linear low-rank (B2)	0.170	0.108	0.276	0.634	0.175
Smoothing, rewired GRN	0.163	0.108	0.260	0.631	0.174
Smoothing, data-kNN graph	0.172	0.109	0.271	0.630	0.175
Smoothing, STRING only	0.250	0.184	0.375	0.679	0.192
iPerturb: smoothing head	0.252	0.179	0.374	0.680	0.191
iPerturb: GCN head (B5)	0.299	0.201	0.422	0.711	0.191
iPerturb: neighbour-mean head	**0.302**	**0.254**	**0.440**	**0.737**	0.204

[+0.030, +0.070] (neighbour mean). The GCN loses its gain on the rewired graph (+0.123 [+0.107, +0.139] and +0.118 [+0.094, +0.141] in favour of the real GRN). We therefore regard the three heads as alternative read-outs of the same context-specific GRN; the smoothing head is the one fixed before evaluation. The selected smoothing settings lie at the edge of the fixed grid (fold 0: k=40, α=0.75; fold 1: k=40, α=0.75; fold 2: k=40, α=0.75), so the reported gain may be a lower bound. The area under the precision–recall curve for differentially expressed genes is reported in Table 1 but is prevalence-dominated: the no-change predictor attains the highest value in RPE1.

3.3 The Hill-form operator is faithful to exact Hill propagation and efficient as a matrix product

Faithfulness. The Chebyshev form approximates the Hill curve to 1.9×10^{-4} at degree 32 and to 2.7×10^{-3} at degree 16 (worst case over $k \in [0.05, 1]$, Figure 3a). Inside the full fixed-point iteration on the curated K562 graph with heterogeneous regulator constants and 200 clamped targets, the largest difference

Table 2. Paired differences in ρ_c. Mean difference with 95% bootstrap interval over unique held-out targets.

Comparison (ρ_c, mean difference [95% CI])	K562	RPE1
GRN vs no graph	+0.096 [+0.082, +0.110]	+0.087 [+0.064, +0.110]
GRN vs rewired GRN	+0.097 [+0.081, +0.113]	+0.089 [+0.066, +0.113]
GRN vs data-kNN graph	+0.091 [+0.076, +0.107]	+0.080 [+0.057, +0.104]
GRN vs rewired (self-entries zeroed)	+0.098 [+0.084, +0.113]	+0.089 [+0.066, +0.113]
iPerturb vs linear (B2)	+0.093 [+0.077, +0.108]	+0.082 [+0.059, +0.106]

of the equilibrium from exact Hill propagation is 5.2×10^{-4} at degree 8 and 3.5×10^{-5} at degree 16 (decoded response up to 2.4×10^{-5}, against a response scale of 0.087); sparse and dense products agree exactly. The SCC-aware solver reproduces the global solver to 1×10^{-13} (largest feedback component 297 of 8,563 genes in K562, 321 of 8,749 in RPE1) with a median speed-up of 2.5× (K562) and 2.2× (RPE1); most of this comes from restricting the computation to the genes reachable from the perturbed gene.

Efficiency. On an A100 (24 damped updates per step), replacing edge-wise gather and scatter by a sparse matrix product is slower on the sparse curated graph (37 ms vs. 72 ms at batch 128) but, on the 247k-edge association graph, reduces the step time from 250 ms to 77 ms and the peak memory from 6949 MB to 1078 MB at batch 128 (960 ms to 208 ms and 27689 MB to 3338 MB at batch 512; Figure 4c). The Chebyshev expansion itself does not speed up a regulator-shared Hill curve with $n = 2$, whose closed form is already cheap (185 ms at degree 16 and batch 128). Its value lies in edge-specific learnable shapes: edge-wise evaluation of per-edge Chebyshev shapes needs activations of size $B \times E \times r$ and runs out of memory on the 40 GB device at batch 128 with $r = 16$ and at batch 512 with $r = 8$, whereas the matrix form stacks a basis of size $B \times G \times r$ and applies one sparse matrix with Er non-zeros, fitting in 5.1 GB and 9.9 GB at batch 128 (Figure 4d; outputs agree to 10^{-7}). On the CPU, the edge-wise form was the fastest or within a few per cent of the fastest in every configuration measured (Appendix B).

Predictive effect. Following the pre-registered gate, we trained the shared-Hill model and a linear model with the same graph, decoder and budget as a residual on the linear baseline for the reachable targets (37 K562 and 50 RPE1 targets over three folds). The Hill form did not improve the centred correlation over the linear form (+0.0013 [0.0000, +0.0037] in K562, +0.0005 [0.0000, +0.0014] in RPE1), over the linear baseline without graph (+0.0003 [-0.0002, +0.0009], -0.0004 [-0.0012, +0.0001]) or over a rewired graph (+0.0002 [-0.0001, +0.0007], -0.0004 [-0.0012, +0.0001]), and an oracle readout with its scale fitted on the test targets themselves added nothing, so the limit is the information in the signed core rather than the optimiser. The Hill-form operator is therefore a faithful

and efficient implementation of the regulatory model, not a source of additional predictive accuracy on this data.

3.4 Predictions are explained by named neighbours and tested by ablation

For each test target we attribute the prediction to its association neighbours by the change in the predicted shift when a neighbour is removed (no observed data enter the attribution) and test the attribution by ablation against the observed profile. Removing the five top-attributed neighbours lowers the centred correlation more than removing five random neighbours by +0.008 [+0.001, +0.016] (K562, 651 targets) and +0.023 [+0.005, +0.041] (RPE1, 689); for a single neighbour the difference is not distinguishable from zero (0.000 [-0.002, +0.003], +0.004 [-0.002, +0.011]). The attributions are therefore weakly but consistently faithful (Figure 5a). The largest gain in the pre-specified K562 fold is DNTTIP2, whose three most attributed neighbours are the ribosomal proteins RPS4X, RPS18 and RPS6; the smallest gain is a target without neighbours, for which the shrunken embedding performs worse than the linear model, and the first neighbour-free target shows a small gain (Figure 5b–d). A fallback to the linear prediction for neighbour-free targets changes the centred correlation by at most 0.002, because such targets are under 2% of the test set.

4 Discussion

We introduced iPerturb, which builds a cell-line-specific regulatory network from five evidence tiers and propagates perturbations through it with a learnable transfer function that follows the Hill function, so that every prediction can be traced to named genes and edges. On held-out CRISPRi targets in K562 and RPE1 the context-specific network carries predictive signal that matched rewired and data-derived graphs do not, and the Hill-form operator is an exact-to-10^{-5}, memory-lean implementation of the regulatory model on accelerators. Its one clear limitation is the one we set out to test: on the curated signed core, which reaches only 5–7% of the perturbed genes in these essential-gene screens, a nonlinear Hill transfer does not predict better than a linear one, and the gain over a linear low-rank model comes from association evidence rather than from curated regulation. We read this as a statement about coverage, not about biology: the screens perturb mostly non-TF genes, and a signed core cannot propagate what it does not reach.

Several limitations qualify these results. The predicted endpoint is the mean response, not a distribution of cells. RNA abundance is a proxy for regulator activity and endpoint data cannot identify time constants, so the fixed-point operator is a steady-state model; neighbour attributions are tested by ablation but are association paths, not validated mechanisms. The folds and seeds reuse the same biological screens, and graph and no-graph models were tuned on comparable but not identical budgets; matched controls share their grid. The

largest gains are obtained for genes in well-characterised complexes (for example, ribosomal-protein neighbours), which are enriched in essential-gene screens, and the transfer to screens with few such complexes is untested. The GeneHancer tier is empty for RPE1, so the cell-line-specific tier is tested in K562 only. Future work will add more cell lines and a genome-wide screen with more transcription-factor targets, and test edge-specific learnable shapes, which the matrix-multiplication form now makes affordable, on data in which a signed core has more reach.

Conflict of interests, acknowledgments, funding, and data availability

Conflict of Interests. The authors declare that they have no competing interests.

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Data and Software Availability. The CRISPRi Perturb-seq datasets analysed in this study are publicly available from Replogle et al. [18]. Prior biological knowledge was obtained from TRRUST, CollecTRI, OmniPath, GeneHancer v5.26, STRING v12 and COXPRESdb; GeneHancer is licence-gated and is not redistributed. The source code, the frozen pre-registration of the evaluation protocol with its later addenda, figure-generation scripts and the files from which every number in this paper is generated are available at https://github.com/kagtgi/iPerturb (branch `ipt-rebuild`).

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A Additional protocol details

Splits and fitting. Three target-held-out folds per line (70/15/15%, stratified by TF status, control-expression tertile and curated out-degree bin; seed 42, fold-specific seeds). The smoothing head selects $k \in \{5, 10, 20, 40\}$ and $\alpha \in \{0, 0.25, 0.5, 0.75\}$ on validation perturbations (selected settings: K562, fold 0: k=40, α=0.75; fold 1: k=40, α=0.75; fold 2: k=40, α=0.75; RPE1, fold 0: k=40, α=0.75; fold 1: k=40, α=0.75; fold 2: k=40, α=0.75); the ridge penalty is 10^{-2}. The GCN has two layers, hidden size 64, residual self-weights, AdamW (learning rate 10^{-3} or 3×10^{-3}, weight decay 10^{-3}), at most 200 epochs, patience 20, three seeds. Hill-form models use $n = 2$, damping 0.5, 24 updates, learning rates $\{0.003, 0.01\}$ and penalties $\{10^{-3}, 10^{-1}\}$ on the per-gene decoder offset, selected on validation reachable targets; the clamp strength ζ is the median training knockdown.

Pre-registration. The evaluation protocol was written and hashed before any graph model was fitted. Later analyses (association smoothing, controls, GCN, explanations, context-specific tier, efficiency study, neighbour-free fallback) were added as dated addenda before they were run; the fallback analysis is exploratory and is not part of the headline numbers.

GEARS. Trained with the official package on at most 100 cells per perturbation and 3,000 control cells, log1p of counts per 10^4, 10 epochs, hidden size 64, with its default gene-ontology graph; targets absent from its ontology set are excluded from its evaluation.

B Efficiency measurements

On the CPU, the edge-wise form was the fastest for the sparse curated graph at every batch size (387 ms at batch 128 against 1485 ms for the sparse product), and on the association graph the dense product matched it (3477 ms vs. 3580 ms).

Table 3. Time per forward and backward step (ms) and peak memory (MB) of the regulator-shared Hill operator (24 updates, A100 40 GB). E0: exact Hill, edge gather/scatter; E1: exact Hill, sparse matrix product; E2: dense product; C1: degree-16 Chebyshev form, sparse product.

Graph	Batch	E0	E1	E2	C1
curated, 10k edges	128	37 / 942	72 / 1054	104 / 1592	179 / 4001
curated, 10k edges	512	73 / 3708	151 / 3315	346 / 3791	305 / 15018
association, 247k edges	128	250 / 6949	77 / 1078	103 / 1606	185 / 4025
association, 247k edges	512	960 / 27689	208 / 3338	346 / 3803	360 / 15041

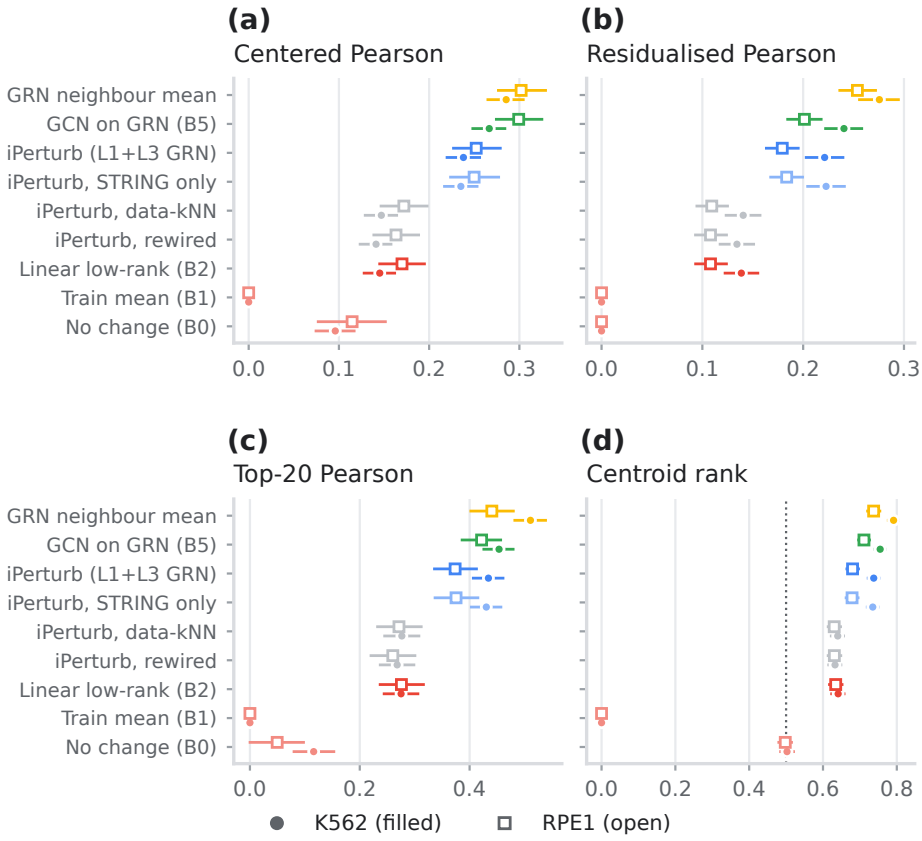


Fig. 2. The context-specific GRN improves response prediction for unseen CRISPRi targets in two human cell lines. Mean over held-out targets with 95% bootstrap intervals over unique targets (three target-held-out folds; K562 $n = 699$, RPE1 $n = 743$). Filled circles, K562; open squares, RPE1. **(a)** Pearson correlation of the response after removing the training-mean response (ρ_c, primary; the train mean scores 0 by construction). **(b)** The same after projecting the training-mean direction out of both profiles (ρ_r). **(c)** Top-20 centred correlation. **(d)** Mean normalised rank of the correct perturbation (centroid accuracy; 0.5 is chance). Smoothing, rewired GRN and data-kNN are controls with the same capacity and tuning grid.

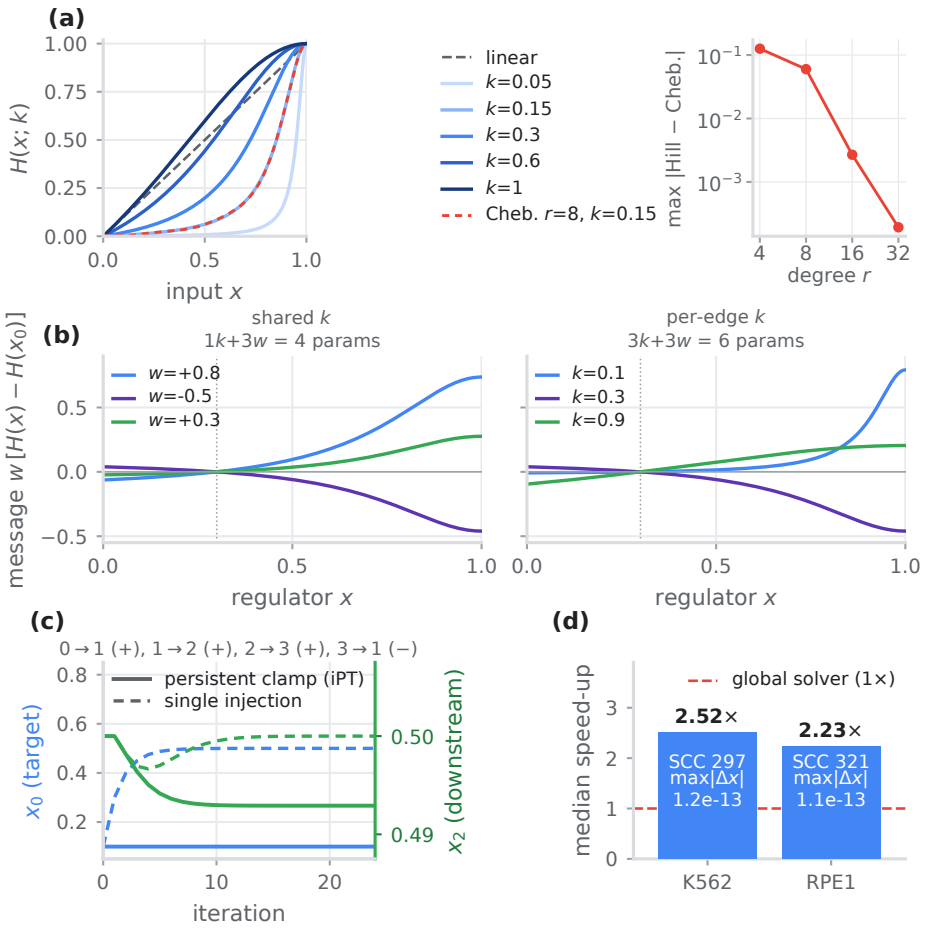


Fig. 3. The regulatory operator. **(a)** Modified Hill transfer functions of Eq. (1) for five dissociation constants, the linear transfer, and the degree-8 Chebyshev form for $k = 0.15$; the inset shows the maximal approximation error against the degree. **(b)** Messages of one regulator to three targets with a shared k (rescaled copies of one shape) or edge-specific k (three shapes). **(c)** A persistent clamp re-imposes the perturbed gene at every update, whereas a single injection lets it relax (four-gene motif with feedback; x_2 on the right axis). **(d)** Median speed-up of the SCC-aware solver over the global solver (numpy reference implementation).

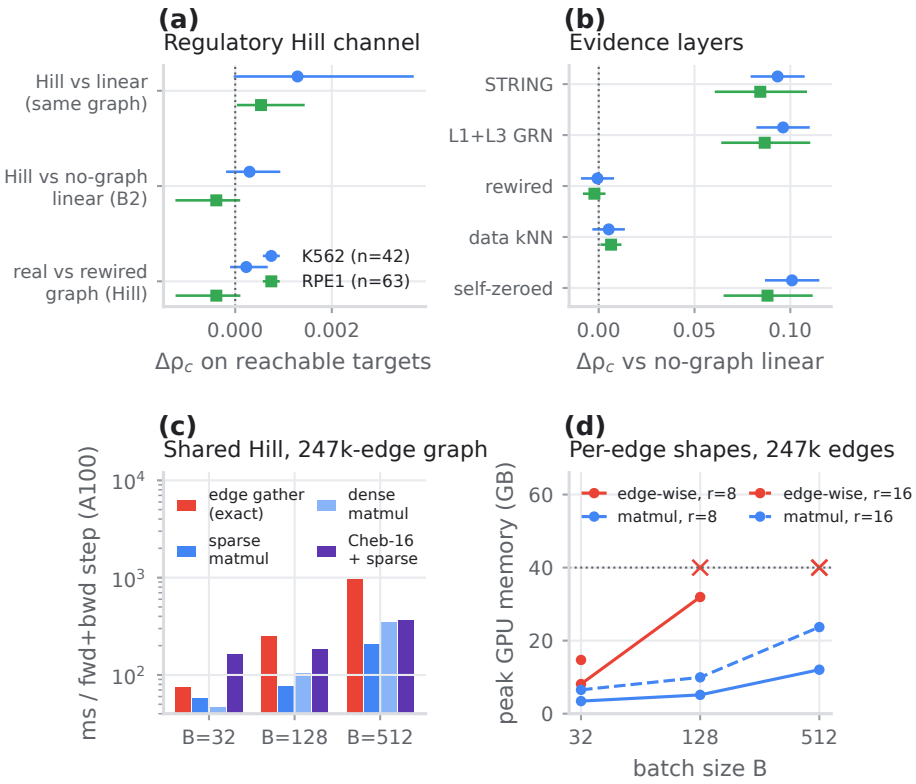


Fig. 4. Ablations and efficiency. **(a)** Hill-form versus linear propagation on the curated signed core, reachable test targets only (differences in ρ_c; 42 K562 and 63 RPE1 test instances). **(b)** Change in ρ_c relative to the no-graph linear model for evidence layers and controls (STRING only; L1+L3; rewired; graph learned from training data; L1+L3 with own-target entries zeroed). **(c)** Time per forward and backward step for the regulator-shared Hill operator on the 247k-edge association graph (A100). **(d)** Peak memory for edge-specific Chebyshev shapes (crosses: out of memory on 40 GB).

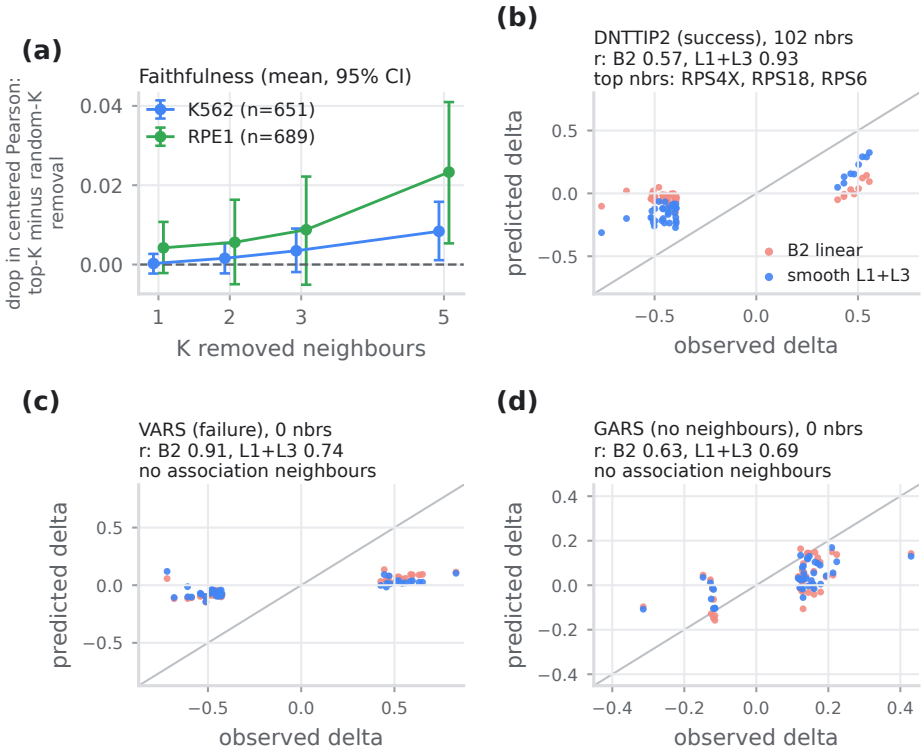


Fig. 5. Explanations of the association channel. (a) Removing the top-K attributed neighbours of a test target lowers its agreement with the observed profile more than removing K random neighbours (mean difference with 95% interval; targets with at least six neighbours). **(b–d)** Cases chosen by a rule fixed beforehand (K562 fold 0): largest gain over the linear baseline, smallest gain, and the first target without neighbours. Neighbours are association evidence, not causal paths.